

MOHAMMAD HABIBUL HASSAN KHAN 22321072

Md. Saiedur Rahman Saied 22261528

LIKHON CHANDRA SARKER 21310007

ABDULLAH AL MAHMUD 22201502

B K M Bodruzzaman 23101426

Name: _____ ID: _____ Sec: 17 Group: 1 Date: _____

Experiment no: 7

Name of the Experiment: DETERMINATION OF THE INTERNAL RESISTANCE
OF A CELL BY USING A POTENTIOMETER

Questions on theory (all diagrams should be drawn by using a pencil and a scale)

*1) What is internal resistance of a cell? [0.25]

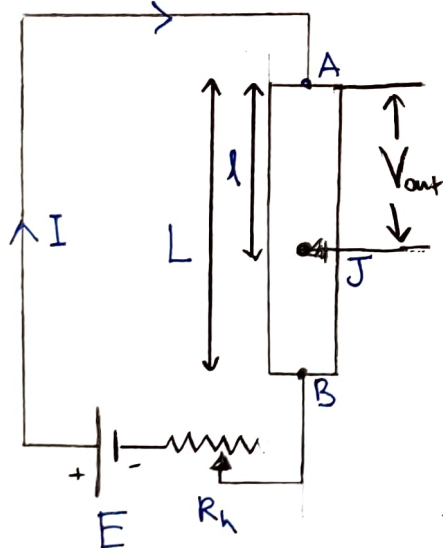
Ans: When the external circuit is complete, within the cell a current flows from the plate at a lower potential to the plate at higher potential and the medium between the plates offers a resistance to the flow of the current.

This resistance is known as the internal resistance of the cell.

It is the resistance in ohms obtained by dividing the difference in volts between generated e.m.f and P.D between the terminals of the cell by the

*2) See Figure 2. Draw it and work out an expression of V_{out} in terms of E , l , L , R , R_h and σ . σ is the resistance per unit length of the conductor AB. R is the total resistance of the conductor. [1]

Ans:



$\Rightarrow$ Resistance's of the conductors segments AJ & JB are R_{AJ} & R_{JB} respectively.

$\Rightarrow$ Total resistance of conductor : $R = R_{AJ} + R_{JB}$

$\Rightarrow$ Length of the conductor is L

$\Rightarrow$ length of the AJ segment of conductor is l

$\Rightarrow$ Conductor is of uniform cross-section, therefore resistance of any segment is proportional to the length of that segment.

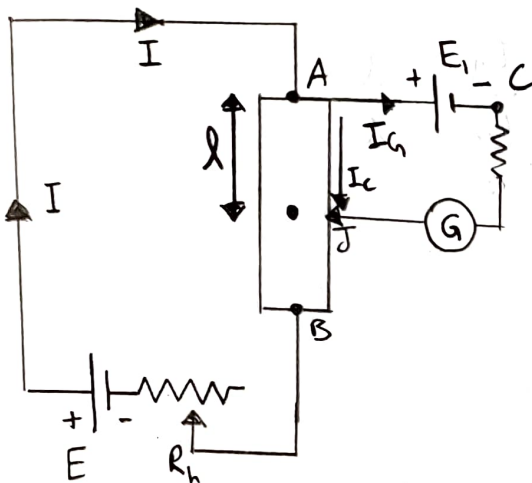
$\Rightarrow \sigma$ is the resistance per unit length of the conductor, $\therefore R_{AJ} = l\sigma$ & $R = L\sigma$

$\Rightarrow$ According to the potential division rule: $V_{out} = \frac{R_{AJ}}{R_{AJ} + (R_{JB} + R_h)} E$

$$= \frac{R_{AJ}}{R + R_h} E = \frac{l\sigma}{L\sigma + R_h} E = \frac{E}{L + \frac{R_h}{\sigma}} l \Rightarrow \left| V_{out} = \frac{E}{L + \frac{R_h}{\sigma}} l \right|$$

*3) See Figure 3. Draw it and work out an expression of the current passing through the galvanometer, I_G as shown in equation (6). [2]

Ans:



$\Rightarrow$ According to Ohm's law,

$$V_C - V_J = I_A R_0 \quad \text{--- (1)}$$

$\Rightarrow$ Voltage difference across E_1 ,
 $V_A - V_C = E_1 \quad \text{--- (2)}$

$\Rightarrow$ By summing eqns (1) & (2)
 $V_A - V_J = E_1 + I_A R_0$

$$\Rightarrow V_A - V_J = V_{out}$$

$$\therefore \frac{E}{L + \frac{R_h}{\sigma}} l = E_1 + I_A R_0$$

$$\Rightarrow \frac{E}{L + \frac{R_h}{\sigma}} l = E_1 + I_A R_0$$

$$\Rightarrow \frac{E}{L + \frac{R_h}{\sigma}} l - E_1 = I_A R_0$$

$$\Rightarrow I_A = \frac{1}{R_0} \left[\frac{E}{L + \frac{R_h}{\sigma}} l - E_1 \right]$$

*4) When I_G is zero, then show that, $E_1 = \frac{E}{L + \frac{R_h}{\sigma}} l$ [0.25]

Ans: $I_A = \frac{1}{R_0} \left[\frac{E}{L + \frac{R_h}{\sigma}} l - E_1 \right]$

if $I_A = 0$

$$0 = \frac{1}{R_0} \left[\frac{E}{L + \frac{R_h}{\sigma}} l - E_1 \right]$$

$$\Rightarrow E_1 = \frac{E}{L + \frac{R_h}{\sigma}} l$$

*5) If l_1 and l_2 are the length of the segment of the potentiometer's wire between end A and the null point,

correspond to E_1 and E_2 , then show that $\frac{E_1}{E_2} = \frac{l_1}{l_2}$ [0.25]

Ans: $\Rightarrow$ For l_1 , $E_1 = \frac{E}{L + \frac{R_h}{\sigma}} l_1$ — (1)

$\Rightarrow$ For l_2 , $E_2 = \frac{E}{L + \frac{R_h}{\sigma}} l_2$ — (2)

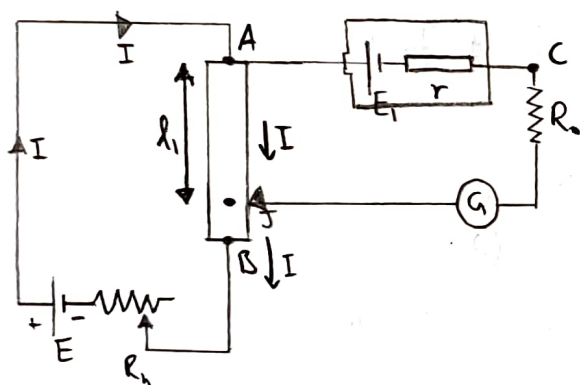
(1) $\div$ (2),

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

*6) Draw the Figure 5 and 7. For Figure 5, show that $E_1 = \frac{E}{L + R_h/\sigma} l_1$ and for Figure 7 show that

$E_1 - Ir = \frac{E}{L + R_h/\sigma} l_2$. Finally show that, $r = \left(\frac{l_1}{l_2} - 1\right)R$. [0.25]

Figure 5:



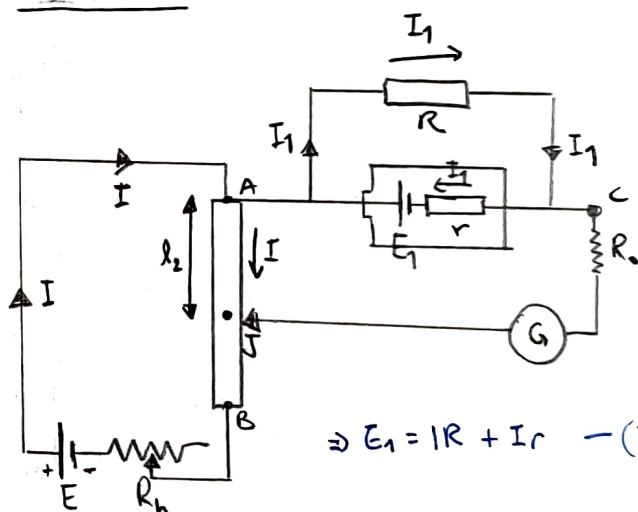
$\Rightarrow$ At the null point, $I_g = 0$

$$\therefore I_g = \frac{1}{R_0} \left[\frac{E}{L + \frac{R_h}{\sigma}} l_1 - E_1 \right]$$

$$\Rightarrow 0 = \frac{1}{R_0} \left[\frac{E}{L + \frac{R_h}{\sigma}} l_1 - E_1 \right]$$

$$\Rightarrow E_1 = \frac{E}{L + \frac{R_h}{\sigma}} l_1 \quad \text{--- (1)}$$

Figure 7:



$$\Rightarrow E_1 = IR + Ir \quad \text{--- (2)}$$

$\Rightarrow$ Voltage difference across R = P.D between the two electrodes of the E_1 , i.e. IR

$$\therefore IR = \frac{E}{L + \frac{R_h}{\sigma}} l_2$$

$$\therefore E_1 - Ir = \frac{E}{L + \frac{R_h}{\sigma}} l_2 \quad \text{--- (3)}$$

$$\Rightarrow (1) \div (3): \frac{E_1 - Ir}{E_1} = \frac{l_2}{l_1}$$

$\Rightarrow$ We know from eqn (2) that $I = E_1/(R+r)$

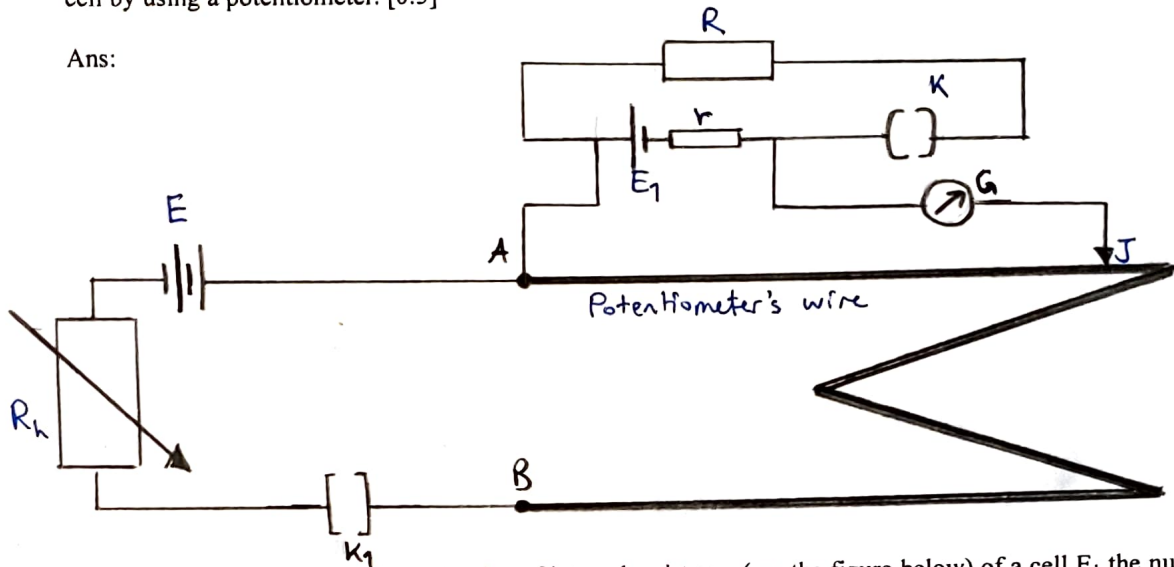
$$\therefore 1 - \frac{r}{R+r} \frac{E_1}{E_1} = \frac{l_2}{l_1} \Rightarrow \frac{R+r-r}{R+r} = \frac{l_2}{l_1}$$

$$\Rightarrow \frac{R+r}{R} = \frac{l_1}{l_2} \Rightarrow 1 + \frac{r}{R} = \frac{l_1}{l_2}$$

$$\Rightarrow r = \left(\frac{l_1}{l_2} - 1\right)R$$

*7) Draw the schematic diagram of the circuit construction to determine the internal resistance of a given cell by using a potentiometer. [0.5]

Ans:



8) In the experiment of the determination of internal resistance (see the figure below) of a cell E_1 the null point of the galvanometer is found at point J_1 when the key K is opened and at the point J_2 when the key K is closed. If the resistance of the resistance box, $R = 5 \text{ Ohm}$ then what is the internal resistance of E_1 ? [0.5]

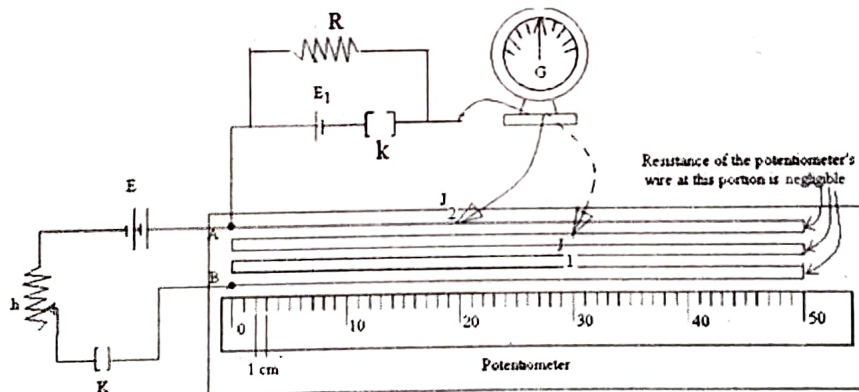


Figure 10: Figure for question (8)

Ans: $l_1 = (50 + 20) \text{ cm} = 70 \text{ cm}$, $l_2 = 20 \text{ cm}$, $R = 5 \Omega$

$$\begin{aligned}
 r &= \left(\frac{l_1}{l_2} - 1 \right) R \\
 &= \left(\frac{70}{20} - 1 \right) (5) \\
 r &= 12.5 \Omega
 \end{aligned}$$

- Draw the data table(s) and write down the variables to be measured shown below (in the 'Data' section), using pencil and ruler BEFORE you go to the lab class.
- Write down your NAME and ID on the top of the page.
- This part should be separated from your Answers of "Questions on Theory" part.
- Keep it with yourself after coming to the lab.

Data

Table: Data for calculating the electromotive force of a cell.

No of Obs.	Circuit	Resistance R (in ohms)	Value of			Internal Resistance r of cell	Mean r In ohms
			l_1 (cm)	Mean l_1 (cm)	l_2 (cm)		
	open	infinity	<u>24.3</u>	<u>243.3</u>			
			<u>24.4</u>				
			<u>24.3</u>				
1.	Closed	10			231	0.532	1.22
2.	Closed	25			232	1.2176	
3.	Closed	30			234	1.2230	
4.	Closed	35			235	1.2361	
5.	Closed	45			237	1.1562	

- READ the PROCEDURE carefully and perform the experiment by YOURSELVES. If you need help to understand any specific point draw attention of the instructors.
- DO NOT PLAGIARIZE data from other group and/or DO NOT hand in your data to other group. It will bring ZERO in this experiment. Repetition of such activities will bring zero for the whole lab.
- Perform calculations by following the PROCEDURE . Show every step in the Calculations section.
- Write down the final result(s)

Calculations
$$r = \left(\frac{l_1}{l_2} - 1 \right) R = \left(\frac{243.3}{237} - 1 \right) 45 = 1.1562$$

Results
$$\bar{r} = \frac{0.532 + 1.2176 + 1.2230 + 1.2361 + 1.1562}{5}$$

$$= 1.22 \Omega$$

$\therefore$ The internal resistance of the cell is 1.22Ω

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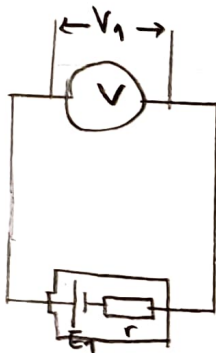
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Questions for Discussions

- 1) Explain a way to determine the internal resistance of a cell by using a voltmeter and a resistance box. Show the derivation of the working formula by assuming the current passing through the voltmeter is negligible. [0.5]

Ans:

⇒ Step 1: measure voltage difference between two electrodes of a real cell when it is not connected with any resistor :-



$$V_1 = E_1 \quad \text{--- (1)}$$

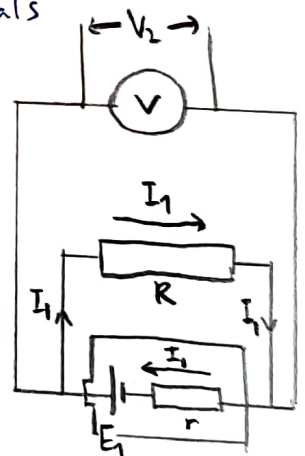
⇒ Step 2: measure the voltage across the two terminals of the cell again when it is connected with a resistor of resistance R.

$$V_2 = \frac{R}{r+R} E_1 \quad \text{--- (2)}$$

$$\Rightarrow \frac{(1)}{(2)}: \frac{V_1}{V_2} = \frac{E_1(r+R)}{E_1 R} \Rightarrow \frac{V_1}{V_2} = \frac{r+R}{R}$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{r}{R} + 1$$

$$\Rightarrow r = \left(\frac{V_1}{V_2} - 1 \right) R$$



- 2) We can use a voltmeter/multi-meter and a resistance box to measure the internal resistance of a cell. What might be a drawback of using a voltmeter or a multi-meter for this purpose? How can a potentiometer be useful to avoid this drawback? [1]

Ans:

- ⇒ Using a voltmeter/M.M to measure internal resistance in reality a tiny amount of current flows through the voltmeter's coil. This current will also flow through the electrolyte(s), inside the cell between its two electrodes. This current inside the cell has to overcome the cell's internal resistance which results a little ~~the~~ potential drop across the cell. Hence V_1 is slightly lower than the actual emf of the cell.
- ⇒ A potentiometer for the same process is more useful for accurate value of emf & internal resistance of a real cell because no current passes through the cell (connected in series with the galvanometer), hence no potential drop occurs across the internal resistance of the cell.

This is the advantage of a potentiometer as a measuring device of electromotive force and as well as internal resistance of a real cell.

- 3) Mention the properties of a good potentiometer for performing this experiment. [0.5]

Ans:

- ① Resistance of a certain portion of the potentiometer's wire is proportional to its length.
- ② Cross-section of the wire is constant everywhere.
- ③ Left & right ends of every segment of wire should coincide with the 0 & 100 cm marks of the meter-scale respectively, otherwise 'end correction' should be made.
- ④ Connections of wires should be tight.
- ⑤ Voltage of E should be constant ~~not~~ and some while measuring E_1, E_2 .
- ⑥ The potentiometer's wire should ~~be~~ consist of a material whose thermal coefficient of resistance is very low.
- ⑦ The wire has significant resistance.